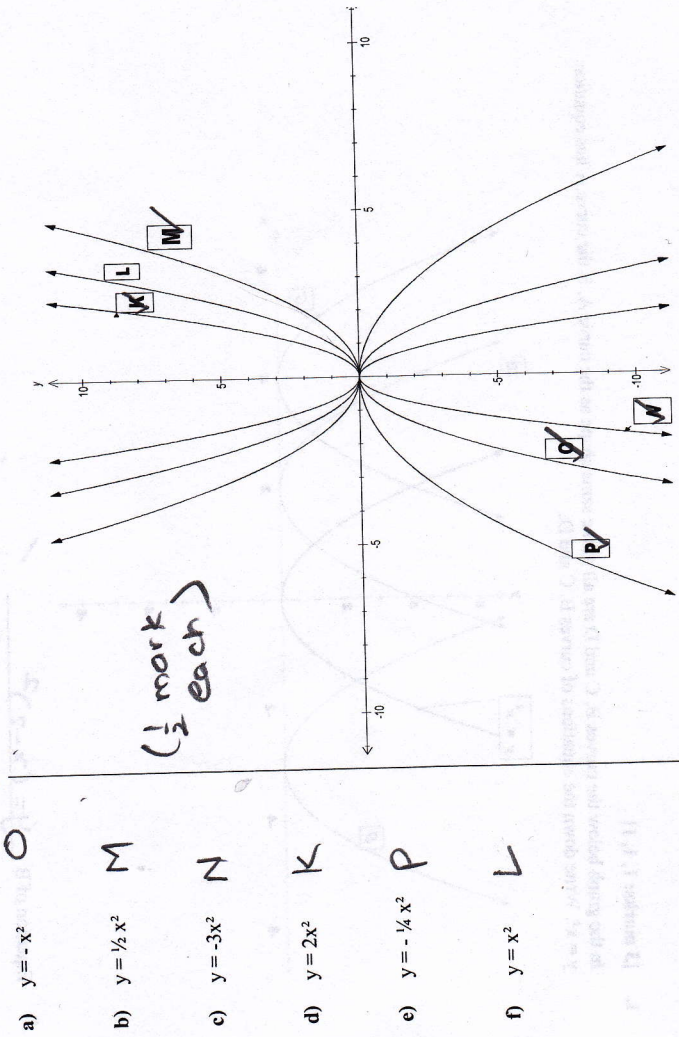


3. [3 marks: $\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}$]

The diagram below shows the graphs of six quadratic functions labelled K, L, M, N, O and P. Match each of the following equations with one of the given graphs



- a) $y = -x^2$ O
b) $y = \frac{1}{2}x^2$ M
c) $y = -3x^2$ N
d) $y = 2x^2$ K
e) $y = -\frac{1}{4}x^2$ P
f) $y = x^2$ L

4. [3 marks: 1, 1, 1]

Describe the following transformation in your own words:

a) $y = x^2 \rightarrow y = (x+1)^2$

The parabola is moved 1 step left (negative x-axis)

b) $y = x^2 \rightarrow y = (x-1)^2 + 2$

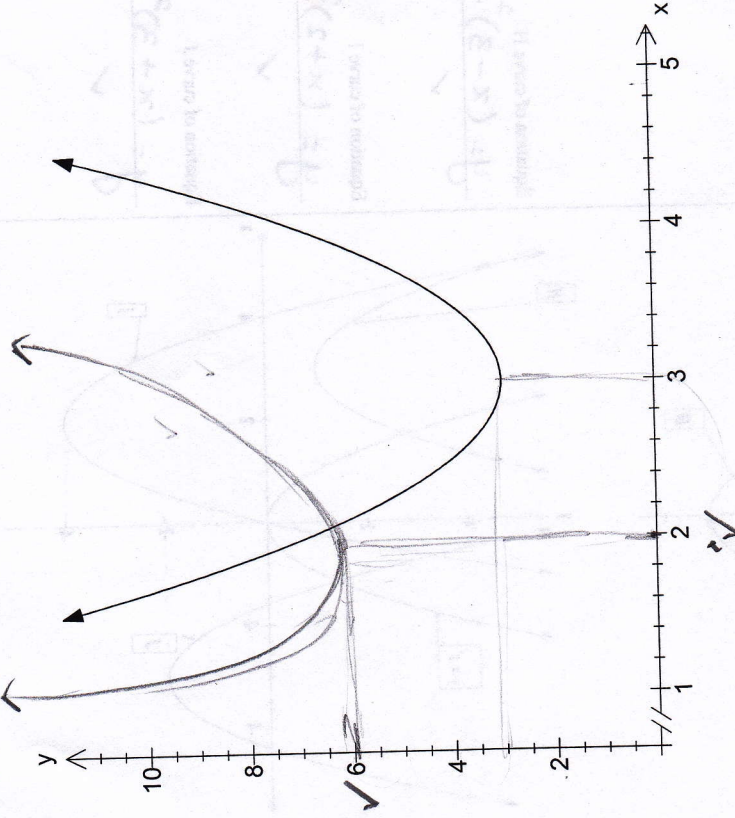
Parabola is moved 1 step right (positive axis) and 2 steps up ($\frac{1}{2}$ mark each) transformation ✓

c) $y = x^2 \rightarrow y = -0.5x^2$

The parabola will be wider and upside down ✓ ($\frac{1}{2}$ mark each) transformation

5. [5 marks: 2, 3]

The following graph shows a quadratic curve with function $y = 4(x-3)^2 + 3$.



- a) Translate the parabola 3 units up and 1 unit left in the same graph.

- b) Write the equation of a new translated parabola.

3 steps up means +3 in the equation will change to +6 ✓

1 unit left means -3 changes to -2. ✓

$y = 4(x-2)^2 + 6$ ✓